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STABLE QUADRATIC POLYNOMIALS OVER $\mathbb{Q}(i)$

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Abstract. We study iterates of a quadratic $f = x^2+1/c \in K[x]$. If the number of factors of $f^n := f\circ f\circ ...\circ f$ is bounded by a constant independent of n, then f is said to be *eventually stable*. This paper is an extension to $\mathbb{Q}(i)$ of the paper [6], which considered f over $\mathbb{Q}$. The conjecture "if f^2 is irreducible, then f^n is irreducible for all n" extends to $\mathbb{Q}(i)$, but the lack of a linear ordering on $\mathbb{Q}(i)$ requires us to pass to the norm. As a result, an auxiliary function is necessary for a lower bound on a finite n to check. The elusive case of $c \equiv 2 \bmod 4$ (as a $\mathbb{Z}$ equivalence class) is shown to be "stable" over $\mathbb{Q}(i)$, offering more evidence for [6, Conjecture 1]. Stability for $c \equiv 1 \bmod 2$ (as a $\mathbb{Z}[i]$ equivalence class) is not as fully handled as over $\mathbb{Z}$, however.

1. Introduction

Suppose K is a field with algebraic closure $\overline{K}$. Let $f(x) = x^2 + r \in K[x]$, and denote its iterates by

$$f^n(x) := (f \circ f \circ ... \circ f)(x) \ (n \text{ compositions}) \qquad \text{with } f^0(x) = x.$$

Fix $\alpha \in K$ and denote by $f^{-n}(\alpha) := \{\beta \in \overline{K} : f^n(\beta) = \alpha\}$. If $f^n(x) - \alpha$ is separable, the disjoint union $T_{f,\infty}(\alpha) := \{\alpha\} \sqcup f^{-1}(\alpha) \sqcup f^{-2}(\alpha) \sqcup \cdots$ is a rooted tree with root vertex α, and edges assigned according to the action of f. Denote by $\mathrm{Aut}(T_f(\alpha))$ the tree automorphisms of $T_f(\alpha)$. Since the natural action of the Galois group $G_K := \mathrm{Gal}(\overline{K}/K)$ on $f^{-n}(\alpha)$ commutes with $f \in K[x]$, we obtain a homomorphism $\mathrm{Gal}(\overline{K}/K) \to \mathrm{Aut}(T_f(\alpha))$ called the arboreal Galois representation associated to (f, α). One of the central problems in arithmetic dynamics is whether the image of this homomorphism has finite index in $\mathrm{Aut}(T_f(\alpha))$; a dynamical parallel of the Serre open image theorem. This question has generated a large literature, see [1] for a summary of this and other current research in the field.

In this article, we attempt to characterize factorizations of $f^n(x) - \alpha$ for $\alpha = 0$.

Definition 1.1. *Let K be a field, $f \in K[x]$, and $\alpha \in K$. For $n \geq 1$ let k_n denote the number of irreducible factors of $f^n(x) - \alpha$ over K. We call the pair (f, α) (resp. f) **eventually stable over** K if there is a constant $C(f, \alpha)$ (resp. $C(f, 0)$) such that for all n,*

$$k_n \leq C(f, \alpha).$$

*The pair (f, α) is called **stable over** K if $C(f, \alpha) = 1$, i.e. $f^n(x) - \alpha$ is irreducible over K for all $n \geq 1$.*

Eventual stability is applied in some finite-index results for certain arboreal Galois representations [3, 4], and is equivalent to a bound independent of n on the number of Galois orbits on $f^{-n}(\alpha)$ [8, Proposition 2.2]. However, this is a straightforward consequence of the image of the arboreal representation associated to (f, α) having finite index, so finite index of the arboreal representation implies eventual stability. The reverse implication is much less clear. Other applications include finiteness of S-integer points in backwards orbits; see [8, Section 3] for a discussion of these and other results concerning the topic.

The following theorem of Hamblen, Jones, and Madhu [7] establishes eventual stability of a large class of polynomials over number fields (using ideas resembling Eisenstein's criterion):

Theorem 1.2. *(Hamblen, Jones and Madhu, [7, Theorem 5]) Let $d \geq 2$, let K be a field of characteristic not dividing d, and let $f(x) = x^d + r \in K[x]$. If there is a discrete non-archimedean absolute value on K with $|r| < 1$, then f is eventually stable over K.*

The following corollary is an extension of [7, Corollary 6] considered by DeMark et al.:

Corollary 1.3. *Let $f(x) = x^d + r \in \mathbb{Q}(i)[x]$, and suppose that r is non-zero and is not the reciprocal of an element of $\mathbb{Z}[i]$. Then f is eventually stable over $\mathbb{Q}(i)$.*

This paper is concerned with the case when $d = 2$ and a non-zero $r \neq -1$ *is* a reciprocal of an element of $\mathbb{Z}[i]$. In this case, "Eisenstein-type" methods break down. The following conjecture attempts to fully describe the situation. It is an extension of [6, Conjecture 1.7] which considered $r \in \mathbb{Q}$, but negatives have turned into squares, while the c-factorization $4m^2(m^2-1)$ for f^2-reducibility over $\mathbb{Z}$ expands to $c = \alpha^2(2i - \alpha^2)$:

Conjecture 1.4. *Let $f_r = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\{0, -1\}$, and let k_n denote the number of irreducible factors of $f_r^n(x)$. Then f_r is eventually stable over $\mathbb{Q}(i)$ with constant $C(f_r, 0) = 4$:*

1. *If $c = \alpha^2$ with $1 \pm i\alpha \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$ then $k_n = 2$ for all $n \geq 1$.*
2. *If $c \in \{\pm 8i, -16\} \subset \mathbb{Z}[i]^2$, then $k_1 = k_2 = 2$, $k_n = 3$ for all $n \geq 3$.*
3. *If $c = (i\sigma^2 - i)^2$ for $\pm\sigma \in \mathbb{Z}[i]\backslash\{3, 5, 56\}$ then $k_1 = 2$ and $k_n = 3$ for all $n \geq 2$.*
4. *If $c = (is^2 - i)^2$ for $\pm s \in \{3, 5, 56\}$ then $k_1 = 2$, $k_2 = 3$ and $k_n = 4$ for all $n \geq 3$.*
5. *If $c \neq 48$ and $c = \alpha^2(2i - \alpha^2)$ for some $\alpha \in \mathbb{Z}[i]$, then $k_1 = 1$ and $k_n = 2$ for all $n \geq 2$.*
6. *If $c = 48$, then $k_1 = 1, k_2 = 2$, and $k_n = 3$ for all $n \geq 3$.*
7. *If $c \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$ and $c \neq \alpha^2(2i - \alpha^2)$ for any $\alpha \in \mathbb{Z}[i]$ (i.e. c is not in an above case), then $k_n = 1$ for all $n \geq 1$.*

All irreducibility statements from here on will be over the field K, and $r = 1/c$ where $c \in \mathbb{Z}[i]$. Case (3) and (4) are where $c = \alpha^2$ with $1 \pm i\alpha \in \mathbb{Z}[i]^2$. Case (7) is precisely the case where $f_r^2(x)$ is irreducible and thus case (7) asserts that if $f_r^2(x)$ is irreducible, then $f_r^n(x)$ is irreducible for all $n \geq 1$. This is the main focus of this paper, and we state this as a $\mathbb{Z}[i]$ analogue of [6, Conjecture 1.8]:

Conjecture 1.5. *Let $f_r(x) = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\{0, -1\}$. If $f_r^2(x)$ is irreducible over K, then f is stable over K.*

The determination of squares in the sequence $\{f_r^n(0)\}$ is of central importance in this study. To this end, we use tools such as the non-linear recurrence relation

$$a_1 = 1, \quad a_n(c) = c^{2^{n-1}-1} + a_{n-1}^2(c) \quad \text{for } n \geq 2$$

describing the numerator of $f_r^n(0) = \frac{a_n}{c^{2^{n-1}}}$ for $n \geq 1$ (note $c^{2^{n-1}}$ is a square when $n \geq 2$).

We prove another irreducibility test involving the a_n sequence.

Lemma 1.6. *Suppose that $c \in \mathbb{Z}[i]\backslash\{0\}$, $r = 1/c$ and f_r^2 is irreducible. Let $a_n = a_n(c)$ be defined as above and set*

$$b_n^{\pm}(c) = i(a_{n-1} \pm \sqrt{a_n}) \in \bar{\mathbb{Q}}. \tag{1}$$

If $b_n^{\pm}(c)$ is not a square in $\mathbb{Z}[i]$ (which holds in particular if $a_n(c)$ is non-square in $\mathbb{Z}[i]$) for all $n \geq 3$, then $f_r^n(x)$ is irreducible for all $n \geq 1$.

It is a $\mathbb{Z}[i]$ analogue of [6, Lemma 1.10], but the lack of a linear ordering on $\mathbb{Z}[i]$ requires that we check that b_n^- is non-square as well. This leads us to the following $\mathbb{Z}[i]$ extension of [6, Conjecture 1.11]:

Conjecture 1.7. *Let $b_n^{\pm}(c)$ be defined as in 1. If $c \in \mathbb{Z}[i]\backslash\{0, -1\}$, then $b_n^{\pm}(c)$ is non-square in $\mathbb{Z}[i]$ for all $n \geq 3$.*

The $\{b_n^{\pm}(c)\}$ square test of Lemma 1.6 allows us to prove a $\mathbb{Z}[i]$ analogue of [6, Theorem 1.3]. This theorem is in some ways weaker than its counterpart in [6] due to the lack of linear ordering on $\mathbb{Z}[i]$, however, it also expands on that theorem:

Theorem 1.8. *Let $f_r(x) = x^2 + r$ with $r = 1/c$. Then f_r^n is irreducible for all $n \geq 1$ if c satisfies one of the following conditions:*

1. *$c, c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, $2 \nmid c$ and $16 \nmid c + 1$;*
2. *$c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, and $c \equiv \pm i \mod a + bi$ with $N(a + bi) \equiv 5 \mod 8$*
3. *$c \in \mathbb{Z}[i]\backslash\{0, -1, -1 \pm 2i\}$ and $v_\pi(c)$ is odd for all primes $\pi \mid c$.*

A notable consequence of item (2) is that for integers $c \equiv 2 \mod 4$, f_r is stable over $\mathbb{Z}[i]$ (and therefore $\mathbb{Z}$). Thus, by [6, Corollary 4.6], $c \not\equiv 0 \mod 4$ implies f_r is stable over $\mathbb{Z}$ (when c is non-square in $\mathbb{Z}$).

We prove another result allowing us to check irreducibility of f^k for some finite k to determine stability of f. The lack of a linear ordering on $\mathbb{Z}[i]$ leads to a new function $\Xi(c)$ for our bound, rather than $\epsilon(c)$ of [6, Theorem 1.4]. This also leads to a smaller denominator $\log\sqrt{1+\frac{1}{|c|}} < \log(1+\frac{1}{\sqrt{|c|}})$:

Theorem 1.9. *Let $f_r(x) = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]$ with $|c| \geq 5$. If f^k is irreducible for*

$$k = 1 + \left\lfloor \log_2\left(1 + \frac{\log 4 + \Xi(|c|)/\sqrt{|c|}}{\log\sqrt{1+1/|c|}}\right)\right\rfloor,$$

then all f^n are irreducible. Here $\Xi(|c|)$ is a monotonically decreasing function on $\mathbb{R}_{\geq 5}$ with

$$\sqrt{2} < \Xi(|c|) \leq \Xi(5) < 6.05 \textit{(see Definition 4.6)}$$

Now, we define the orbit of $t \in \mathbb{Q}(i)$ under f_r to be the set $O_{f_r}(t) = \{t, f_r(t), f_r^2(t), ...\}$, and we say that a prime π divides $O_{f_r}(t)$ if there is at least one nonzero $\alpha \in O_{f_r}(t)$ with $v_\pi(\alpha) > 0$. We prove the following $\mathbb{Z}[i]$ analogue of [6, Proposition 6.1], showing that the natural density of prime divisors of orbits is 0 when $b_n^{\pm}$ is non-square for all $n \geq 3$. The difficulty in a proof of this over $\mathbb{Q}(i)$ is constructing the correct tower of Galois extensions on which to apply Chebotarev's theorem:

Theorem 1.10. *Let $f_r(x) = x^2 + r$ for $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\{0, -1\}$, with $c, c+1$ both non-square in $\mathbb{Z}[i]$. If $\{|b_n^+|^2, |b_n^-|^2\}$ contains no squares in $\mathbb{Z}$ for all $n \geq 3$, then for any $t \in \mathbb{Q}(i)$ we have*

$$D(\{\pi \text{ prime in } \mathbb{Z}[i] : \pi \text{ divides } O_{f_r}(t)\}) = 0.$$

2. Background

We first list some work-saving facts:

Proposition 2.1. *If $p(x) \in \mathbb{Q}(i)[x]$, then any factorization of p into irreducible polynomials over $\mathbb{Q}(i)$ has the same number of factors as $\bar{p}(x)$, $p(x)$ with conjugated coefficients. In particular, the polynomial $f_r^n(x)$ has the same number of factors irreducible over $\mathbb{Q}(i)$ as $f_{\bar{r}}^n(x)$.*

Proof. Since conjugation is a field automorphism of $\mathbb{Q}(i)$, it extends to a ring automorphism of $\mathbb{Q}(i)[z]$. Ring automorphisms preserve irreducibility. □

This allows us to only consider c in quadrants 1 and 3, reducing our work load by "half".

We also will need some way of inferring irreducibility of iterated polynomials: the following (derived from Capelli's Lemma) translates irreducibility of the list $\{g(f^n(x))\}_{n\geq 1}$ to a property of a sequence in K.

Lemma 2.2. *(Demark-Hindes-Jones-Misplon-Stoll-Stoneman, [6, Lemma 2.2]) Let K be a field with $char(K) \neq 2$, $g \in K[x]$ a monic, irreducible polynomial with $d := \deg(g) \geq 1$, and $f(x)$ monic and quadratic with critical point β. If the set*

$$\{(-1)^d g(f(\beta))\} \cup \{g(f^n(\beta))\}_{n\geq 2}$$

contains no squares in K, then $g(f^n(x))$ is irreducible over K for all $n \geq 1$.

Remark. *The proof found in [6, Lemma 2.2] shows that if $g(f^n(x))$ is irreducible (over K) for some $n \geq 1$ and $g(f^{n+1}(\beta))$ is non-square in K, then $g(f^{n+1}(x))$ is irreducible (over K).*

When $g(x) = f_r(x) = x^2 + r$, the hypothesis asks if the set $\{f_r^n(0) = \frac{a_n(c)}{c^{2^{n-1}}}\}_{n\geq 1}$ contains no squares in K $\iff$ the set $\{a_n = a_{n-1}^2(c) + c^{2^{n-1}-1}\}_{n\geq 1}$ contains no squares in $\mathbb{Z}[i]$.

This idea that we may infer irreducibility of f^n by considering whether $f^n(0)$ is a square leads to necessary and sufficient conditions for the irreducibility of f_r and f_r^2 when f_r is irreducible. The following specialization is one of the first instances where the lack of ordering changes our results:

Proposition 2.3. *Let $f_r(x) = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\{0\}$. Then $f_r(x)$ is reducible if and only if $c = \alpha^2$ for some $\alpha \in \mathbb{Z}[i]$. If $f_r(x)$ is irreducible, then $f_r^2(x)$ is reducible if and only if $c = \alpha^2(2i - \alpha^2)$ for some $\alpha \in \mathbb{Z}[i]$.*

Proof. The first statement is clear. Now suppose f_r is irreducible over $\mathbb{Q}(i)$. If $c = \alpha^2(2i - \alpha^2)$, then

$$f_r^2(x) = \left(x^2 - \frac{1+i}{\alpha}x - \frac{i\alpha^2+1}{\alpha^2(2i-\alpha^2)}\right)\left(x^2 + \frac{1+i}{\alpha}x - \frac{i\alpha^2+1}{\alpha^2(2i-\alpha^2)}\right).$$

These factors are reducible if and only if their discriminant $\frac{-2i}{\alpha^2-2i}$ is square in $\mathbb{Q}(i)$ (i.e. $\alpha^2 - 2i$ is a square in $\mathbb{Z}[i]$): suppose $\alpha^2 - 2i = \eta^2$. Then $2i = (\alpha - \eta)(\alpha + \eta)$. If $\{\alpha + \eta, \alpha - \eta\} = \{-1 + i, 1 - i\}$, this implies $2\alpha = 0$. Since $c \neq 0$, we have a contradiction. We obtain similar contradictions when $\{\alpha + \eta, \alpha - \eta\} \in \pm\{\{1 + i, 1 + i\}, \{1, 2i\}, \{i, 2\}\}$. Thus the factors of $f_r^2(x)$ are irreducible over $\mathbb{Q}(i)$ when $c = \alpha^2(2i - \alpha^2)$.

The converse is analogous to the proof of [6, Proposition 2.1], but $2(k^2-1)(-1\pm k)$ is potentially a square in $\mathbb{Z}[i]$ instead of only $2(k^2 - 1)(-1 + k)$ over $\mathbb{Q}$ (where $c = k^2 - 1$). We also need to factor $2 = i(1 - i)^2$. □

Non-squares in $K = \mathbb{Q}(i)$ are thus of great interest, so squares in $\mathbb{Z}[i]$ are too. A tactic is checking for quadratic non-residues: if the Gaussian integer α is non-square mod π, then α is non-square in $\mathbb{Z}[i]$. In the rest of this paper, $\left[\frac{\cdot}{\alpha}\right]$ and $\left[\frac{\cdot}{\pi}\right]_2$ refer to the Gaussian integer Jacobi and Legendre symbols (respectively) as in [5]. Thus if $\alpha = \pi_1^{a_1} \cdot \pi_2^{a_2} \cdots \pi_n^{a_n}$ is a prime factorization of $\alpha \in \mathbb{Z}[i]$, then

$$\left[\frac{\sigma}{\alpha}\right] = \left[\frac{\sigma}{\pi_1}\right]_2^{a_1} \cdot \left[\frac{\sigma}{\pi_2}\right]_2^{a_2} \cdots \left[\frac{\sigma}{\pi_n}\right]_2^{a_n} = -1 \quad \text{if } \sigma \text{ is a non-residue } \mod \alpha.$$

The notion of a "rigid divisibility sequence" will also be advantageous in this search for non-squares.

2.1. **Rigid divisibility.** One can show inductively that $f_{1/c}^n(0) = a_n/c^{2^{n-1}}$, where

$$a_1(c) = 1, \quad a_n(c) = a_{n-1}^2(c) + c^{2^{n-1}-1} \quad \text{for } n \geq 2.$$

Note that the numerator a_n is relatively prime to the denominator $c^{2^{n-1}}$, a square when $n > 1$. This numerator inherits prime divisors in a predictable way, leading to the notion of "rigid divisibility":

Definition 2.4. *Let $A = \{z_n\}_{n\geq 1}$ be a sequence in a field K. We say A is a **rigid divisibility sequence over** K if for each non-archimedean absolute value $|\cdot|$ on K, the following hold:*

(1) if $|z_n| < 1$, then $|z_n| = |z_{kn}|$ for any $k \geq 1$.

(2) if $|z_n| < 1$ and $|z_j| < 1$, then $|z_{gcd(n,j)}| < 1$.

The following theorem establishes rigid divisibility of $\{f^n(0)\}_{n\geq 1}$, and implies that the $\{a_n\}_{n\geq 1}$ sequence also has this property:

Theorem 2.5. *(Hamblen, Jones and Madhu, [7, Lemma 12]) Let K be a field and $f(x) = x^d + r \in K[x]$ for some $d \geq 2$. Then $\{f^n(0)\}_{n\geq 1}$ is a rigid divisibility sequence over K.*

The notion of rigid divisibility leads to tables of congruences for c ensuring $f_{1/c}^n$ is irreducible for all n, the $\mathbb{Z}[i]$ analogue of [6, Proposition 3.5]:

Proposition 2.6. *Suppose that $c \in \mathbb{Z}[i]\backslash\{0\}$. If c satisfies any of the congruences in Table 1, or $c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$ and c satisfies any of the congruences in Table 2, then a_n is not a square in $\mathbb{Z}[i]$ for all $n \geq 2$.*

Table 1: Congruences that ensure a_n is non-square for $n \geq 2$

$c \equiv i,\ 1 + i$	$(\mathrm{mod}\ 2)$
$c \equiv -i$	$(\mathrm{mod}\ 2 + i)$
$c \equiv -1 - i$	$(\mathrm{mod}\ 3 + i)$
$c \equiv i,\ \pm 1 + i$	$(\mathrm{mod}\ 3 + 2i)$
$c \equiv 1 + 2i$	$(\mathrm{mod}\ 4)$

Table 2: Congruences that ensure a_n is non-square for $n \geq 2$, provided that $c+1$ is non-square.

$c \equiv i$	$(\text{mod } 2+i)$
$c \equiv -1 \pm i$	$(\text{mod } 3)$
$c \equiv \pm i,\ \pm 1+i,\ 1-i$	$(\text{mod } 3+i)$
$c \equiv -i,\ -2,\ -2i$	$(\text{mod } 3+2i)$
$c \equiv -1+2i,\ \pm 1-2i$	$(\text{mod } 4)$
$c \equiv -2,\ 1+i,\ 1+2i,\ 2-i$	$(\text{mod } 4+i)$
$c \equiv \pm i,\ \pm 2,\ \pm 3,\ \pm(1+i),\ \pm(1-2i),$	$(\text{mod } 4+3i)$
$c \equiv 1,\ 3$	$(\text{mod } 4+4i)$
$c \equiv \pm(2-2i),\ \pm(1+i),\ \pm(1-2i)$	$(\text{mod } 5)$
$c \equiv -2, -2 \pm i, \pm 1-i, \pm i, 2i, 1 \pm 2i, \pm(2+2i)$	$(\text{mod } 5+i)$

Proof. Let $f(x) = x^2 + 1/c$. The sequence $\{f^n(0) = \frac{a_n(c)}{c^{2^{n-1}}}\}_{n>1}$ contains no squares in $\mathbb{Q}(i)$ if and only if the sequence $\{a_n(c)\}_{n>1}$ contains no squares in $\mathbb{Z}[i]$. When $a_n(c)$ is not a quadratic residue mod π, it must be non-square in $\mathbb{Z}[i]$. Table 1 consists of such pairs z, π with $c \equiv z \mod \pi$ implying $a_n(c)$ is a non-residue mod π for all $n \geq 2$: for example, when $c \equiv i \mod 2$, the relationship $a_k(c) = c^{2^{k-1}-1} + a_{k-1}(c)^2$ yields

$$\{1, a_2 = c+1, a_3, ...\} \equiv \{1, 1+i, i, 1+i, ...\} \text{ mod } 2.$$

Since $\left[\frac{i}{2}\right]_2 = \left[\frac{1+i}{2}\right]_2 = -1$ (i.e. i and $1+i$ are not quadratic residues mod 2), a_k is non-square in $\mathbb{Z}[i]$ for all $k \geq 1$.

We now show that a_p non-square implies a_{kp} is non-square for $c \in \{0,1\}$ mod 2 (since $c \in \{i, 1+i\}$ mod 2 are already handled). By rigid divisibility, $v_\pi(a_p) = v_\pi(a_{kp})$, so we can conclude that a_{kp} is non-square in $\mathbb{Z}[i]$ if we know that $v_\pi(a_p)$ is odd.

When $c \equiv 0 \text{ mod } 2$, and p is an integer prime, a_p either has a prime divisor of odd multiplicity or is of the form $i\alpha^2$ when it is non-square. To rule out the latter, note that $a_1(c) = 1$ and for all $n \geq 2$, $a_n = a_{n-1}^2 + c^{2^{n-1}-1} \equiv 1 + 0 \equiv 1 \text{ mod } 2$. However, $i\alpha^2 \in \{0, i\} \text{ mod } 2$ when $\alpha \in \mathbb{Z}[i]$. Since $a_n \equiv 1 \text{ mod } 2$ for all n, a_p and thus a_{kp} has this prime divisor to the same multiplicity, hence is non-square.

Now suppose $c \equiv 1 \text{ mod } 2$. To show that $a_p \neq i\alpha^2$, we start with $p = 2$. Then $a_2 = c+1 \equiv 0 \text{ mod } 2 \implies a_{2n} \equiv 0 \text{ mod } 2$ by rigid divisibility. We show a_{2n} is non-square when a_2 is non-square: if $a_2 = c+1$ has some prime divisor π of odd multiplicity, rigid divisibility then implies that a_{2n} is non-square for all $k \geq 1$. Now suppose $c+1 = i\alpha^2$. Since $a_{2n}(c) = (c+1)P_{2n}(c)$ over $\mathbb{Z}$, the only way $a_{2n}(c)$ is a square in $\mathbb{Z}[i]$ is if $P_{2n}(c) = i\alpha_1^2$ for some $\alpha_1 \in \mathbb{Z}[i]$. However, by rigid divisibility $v_{1+i}(a_{2n}) = v_{1+i}(c+1)$, so $1+i \nmid P_{2n}(c)$. Since $P_{2n}(c) \in \mathbb{Z}[c]$, $1+i \nmid P_{2n}(c)$, and $c \equiv 1 \text{ mod } 2$, we must have $P_{2n}(c) \equiv 1 \text{ mod } 2$. Thus $P_{2n}(c) \neq i\alpha_1^2$, so $a_{2n}(c)$ is non-square.

Also, $a_{2n+1} \equiv c^{2^{2n}-1} + a_{2n}^2 \equiv 1 \text{ mod } 2$, so if a_{2n+1} is non-square in $\mathbb{Z}[i]$, $a_{2n+1} \neq i\alpha^2$. We have thus shown $a_p(c)$ non-square implies $a_{kp}(c)$ is non-square when $c \in \{0,1\} \mod 2$.

The rest proceeds analogously to the proof of [6, Proposition 3.5].

□

Remark. *Since $c \equiv i \mod 2$ and $c \equiv 1+i \mod 2$ are in table 1, f_r is stable when f_r^2 is irreducible for such c. We may assume $c \in \{0,1\} \mod 2$ for the rest of the paper.*

The proof of [6, Proposition 3.3] is also easily adapted to the Gaussian integer setting, and rigid divisibility extends this result to the claim that a_{3n} is non-square in $\mathbb{Z}[i]$ for all $n \geq 1$:

Proposition 2.7. *If $c \in \mathbb{Z}[i]\backslash\{0, -1, \pm 2i, -89\}$, then a_3 is non-square in $\mathbb{Z}[i]$.*

Proof. We have $a_3(c) = c^3 + c^2 + 2c + 1$, and so if $a_3(c) = y_0^2$ for $y_0 \in \mathbb{Z}[i]$, then necessarily (c, y_0) is an integer point on the elliptic curve $y^2 = x^3 + x^2 + 2x + 1$. This curve has conductor norm 2116 with label 2.0.4.1-2116.1-b2 in the LMFDB [10]. It has Mordell-Weil group generator $(-1 : i : 1)$ and torsion isomorphic to $\mathbb{Z}/3\mathbb{Z}$ with generator $(0 : -1 : 1)$. We determine all $\mathbb{Z}[i]$-integral points using MAGMA [2] with code from Thongjunthug's thesis [11], :

$$\{(x, \pm y)\} = \{(-1, i), (0, 1), (-2i, 1 + 2i), (2i, 1 - 2i), (-89, 835i)\}$$

Note $c \in \{0, -1, \pm 2i, -89\}$ is excluded by hypothesis. □

3. The proof of Theorem 1.8

The Gaussian integer analogue of Lemma 3.2 of [6] is another important instance of the lack of a total ordering on $\mathbb{Z}[i]$ coming into play, with an identical proof (until the last line):

Lemma 3.1. *Suppose that $r = 1/c$ and $f_r^2(x)$ is irreducible. Let $a_n = a_n(c)$ and*

$$b_n^{\pm}(c) = i(a_{n-1} \pm \sqrt{a_n}).$$

If for every $n \geq 3$, $\{b_n^+(c), b_n^-(c)\}$ contains no squares in $\mathbb{Z}[i]$ (which holds if a_n is non-square in $\mathbb{Z}[i]$), then $f_r^n(x)$ is irreducible for all $n \geq 1$.

Remark. *The proof of the above* [6, Lemma 3.2] *shows that if $f_{1/c}^n(x)$ is irreducible and the set $\{b_{n+1}^+(c), b_{n+1}^-(c)\}$ contains no squares in $\mathbb{Z}[i]$, then $f_{1/c}^{n+1}(x)$ is irreducible for $n \geq 2$.*

The following is a $\mathbb{Z}[i]$ analogue of [6, Theorem 3.6] giving simple criteria for $b_n^{\pm}(c)$ to be non-square for all $n \geq 3$:

Theorem 3.2. *Let $f_r(x) = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\{0\}$. Let a_n and $b_n^{\pm}$ as in equation (1). Assume that c satisfies one of the following conditions:*

1. *$c, c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, $2 \nmid c$ and $16 \nmid c + 1$;*
2. *$c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, and $c \equiv \pm i \bmod a + bi$ with $N(a + bi) \equiv 5 \mod 8$;*
3. *$c \in \mathbb{Z}[i]\backslash\{0, -1, -1 \pm 2i\}$ and $v_\pi(c)$ is odd for all primes π dividing c.*

In item (2), a_n is non-square for $n \geq 2$, while in items (1) and (3), $\{b_n^+(c), b_n^-(c)\}$ contains no squares in $\mathbb{Z}[i]$ for all $n \geq 3$. In all cases, f_r^n is irreducible for all $n \geq 1$.

We break this theorem into three lemmas.

3.0.1. *Case (1).*

Lemma 3.3. *Suppose $c, c + 1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$. If $2 \nmid c$ and $16 \nmid c + 1$, then f_r^n is irreducible for all $n \geq 1$.*

Proof. By Table 1, we may assume $c \notin \{i, 1+i\} \bmod 2$. Note f is irreducible since $c \notin \mathbb{Z}[i]^2$ by Proposition 2.3 and $a_2 = c + 1$ is non-square in $\mathbb{Z}[i]$. By the remark after Lemma 2.2, we have that f_r^2 is irreducible.

Since $a_2 = c + 1$ is non-square by hypothesis and $c \in \{0, 1\} \bmod 2$, note a_{2n} (and hence $b_{2n}^{\pm}$) are also non-square by the proof of Proposition 2.6. We need only show $b_{2n+1}^{\pm}$ is non-square in $\mathbb{Z}[i]$.

Since $2 \nmid c$ and $c \notin \{i, 1 + i\} \bmod 2$, $c \equiv 1 \bmod 2$. By the recurrence relation $a_k = a_{k-1}^2 + c^{2^{k-1}-1}$, we must have $a_k \in \{0, 1\} \bmod 2$ for all $k \geq 1$. If a_k is non-square then $b_k^{\pm}$ is as well, so suppose a_k is square. If $\sqrt{a_k} \in \{0, 1\} \bmod 2$, then

$$\sqrt{a_k} \equiv a_k \equiv a_{k-1}^2 + c^{2^{k-1}-1} \equiv a_{k-1} + 1 \bmod 2.$$

Now,

$$b_k^{\pm} \equiv i(a_{k-1} \pm \sqrt{a_k}) \equiv i(a_{k-1} + (a_{k-1} + 1)) \equiv i \bmod 2,$$

so $b_k^{\pm}$ is non-square modulo 2, hence is non-square in $\mathbb{Z}[i]$. We now suppose $\sqrt{a_k} \in \{i, 1 + i\} \bmod 2$.

Note $c \equiv 1 \bmod 2$ implies $c \in \{\pm 1, \pm 1+2i\} \bmod 4$ and $c+1 \in \{0, 2, 2i, 2+2i\} \bmod 4$. Then $c^2 \equiv 1 \bmod 4$, so $a_k = c^{2^{k-1}-1} + a_{k-1}^2 \equiv c + a_{k-1}^2 \bmod 4$. By rigid divisibility of $\{a_k\}_{k\geq 1}$, we have $2 \mid c+1 \implies 2 \mid a_{2k} \implies 4 \mid a_{2k}^2$, so combining $a_k \equiv c + a_{k-1}^2 \bmod 4$ with this fact yields

$$\{a_k\}_{k\geq 1} \equiv \{1, c+1, c, c+1, c, ...\} \bmod 4 \tag{2}$$

First suppose $\sqrt{a_k} \equiv 1 + i \bmod 2$, so $a_k \equiv 2i \bmod 4$. Since $c \equiv 1 \bmod 2$, $c + 1 \equiv a_{2n} \equiv 2i \bmod 4$ by equation (2). Thus $c \equiv -1+2i \equiv a_{2n+1} \bmod 4$, so c and a_{2n+1} are non-square as well. Thus a_k is non-square for all $k \geq 2$ when $\sqrt{a_k} \equiv 1 + i \bmod 2$.

Now suppose $\sqrt{a_k} \equiv i \bmod 2$. Since $\sqrt{a_k} = 2z + i$, we have $a_k \equiv -1 \bmod 4$. Since $c + 1 \equiv 0 \bmod 2$, equation (2) implies $c \equiv a_k \equiv -1 \bmod 4$. Then $c \in \{-1, -1 + 4i, 3, 3 + 4i\} \bmod 8$, so $c + 1 \in \{0, 4i, 4, 4 + 4i\} \bmod 8$. Again

$$\{a_k\}_{k\geq 1} \equiv \{1, c+1, c, c+1, c, ...\} \bmod 8 \tag{3}$$

by analogous reasoning to equation (2).

If $c \in \{-1 + 4i, 3\} \bmod 8$ we're done, these are non-square and would imply a_{2k+1} is non-square in $\mathbb{Z}[i]$ by equation (3), thus $b_{2k+1}^{\pm}(c)$ is not a square in $\mathbb{Z}[i]$.

If $c \equiv 3 + 4i \equiv (2 + i)^2 \bmod 8$, then $a_{2k+1} \equiv 3 + 4i \bmod 8$ by equation (3) and

$$\sqrt{a_{2k+1}} \in \{-2 - i, -2 + 3i, 2 + i, 2 - 3i\} \bmod 8$$

when a_{2k+1} is a square in $\mathbb{Z}[i]$. Also, $c + 1 \equiv a_{2k} \equiv 4 + 4i \bmod 8$ and thus

$$b_{2k+1}^{\pm}(c) \equiv i(4 + 4i \pm \sqrt{3 + 4i}) \in \{-3 + 2i, -7 + 2i, -5 + 6i, -1 + 6i\} \bmod 8.$$

Since this set consists of non-squares mod 8, $b_{2k+1}^{\pm}$ is not a square in $\mathbb{Z}[i]$ when $c \not\equiv -1 \bmod 8$.

If $c \equiv -1 \bmod 8$, then $c \equiv a_{2k+1} \bmod 8$ and $a_{2k} \equiv c + 1 \equiv 0 \bmod 8$ by equation (3). Therefore $\sqrt{a_{2k+1}} \in \{\pm i, \pm 3i\} \bmod 8$. If $\sqrt{a_{2k+1}} \equiv \pm 3i \bmod 8$, then we have

$$b_{2k+1}^{\pm}(c) \equiv i(a_{2k} \pm \sqrt{a_{2k+1}}) \equiv \pm 3 \bmod 8$$

and is thus non-square. Thus we may assume $\sqrt{a_{2k+1}} \equiv \pm i \bmod 8 \implies a_{2k+1} \equiv -1 \bmod 16$. Also,

$$\{a_k\}_{k\geq 1} \equiv \{1, c+1, c, c+1, ...\} \bmod 16$$

again by the same reasoning as equation (2). Thus $c \equiv a_{2k+1} \equiv -1 \bmod 16$.

Now, when $c, c + 1 \in \mathbb{Z}[i] \backslash \mathbb{Z}[i]^2$, and $2 \nmid c$, we find that $c \not\equiv -1 \bmod 16$ must land into an above case. Thus $\{b_n^+(c), b_n^-(c)\}$ contains no squares in $\mathbb{Z}[i]$ for each $n \geq 3$. Since f_r^2 is irreducible, f_r^n is irreducible for all $n \geq 1$ by Lemma 2.2.

□

Remark. *If $c \equiv -1 \bmod 16$, then $\sqrt{a_{2k+1}} \in \pm\{i, 7i\} \bmod 16$ with*

$$b_{2k+1}^{+}(c) \equiv i(a_{2k} + \sqrt{a_{2k+1}}) \equiv \begin{cases} i(0 \pm 7i) \equiv \pm 7 \\ i(0 \pm i) \equiv \pm 1 \end{cases} \bmod 16$$

(the same is true of b_{2k+1}^{-}) but ± 1 and ± 7 are all squares mod 16: note $-1 = i^2$, while $7 \equiv (3i)^2 \bmod 16$. Thus the method of proof of Lemma 3.3 cannot conclude that $b_{2k+1}^{\pm}(c)$ is non-square for such c.

3.0.2. *Case (2).*

Lemma 3.4. *If $c+1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, and $c \equiv \pm i \bmod a+bi$ with $N(a+bi) \equiv 5 \mod 8$, then $\{a_n\}_{n\geq 2}$ contains no squares in $\mathbb{Z}[i]$. Thus f_r^n is irreducible for all $n \geq 1$.*

Proof. Let $f_r(x) = x^2 + r$ with $r = 1/c$ for $c \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$. If $c \equiv i \bmod a+bi$ then

$$\{a_n(c) \bmod a+bi\}_{n\geq 1} \equiv \{1, 1+i, i, -1-i, i, -1-i, ...\}. \tag{4}$$

Note that $\left[\frac{\pm i}{a+bi}\right] = (-1)^{(a^2+b^2-1)/4}$ when a is odd, b is even by [5, Theorem 17]. I claim the same is true when a and b are reversed: note that

$$\begin{aligned} i = \gamma^2 + z(a+bi) \iff -i &= \bar{\gamma}^2 + \bar{z}(a-bi) \\ &= \bar{\gamma}^2 + \bar{z}(-i\cdot i)(a-bi) \\ &= \bar{\gamma}^2 - i\bar{z}(b+ai), \end{aligned}$$

so

$$\left[\frac{\pm i}{a+bi}\right] = \left[\frac{\pm i}{b+ai}\right] = (-1)^{(a^2+b^2-1)/4} \tag{5}$$

when a and b have opposite parity.

Thus $N(a+bi) = a^2+b^2 \equiv 5 \mod 8$ implies $\left[\frac{\pm i}{a+bi}\right] = -1$ by equation (5). Thus $a_{2k+1} \equiv i \bmod a+bi$ is non-square for all $k \geq 1$ by equation (4).

Since $c+1 = a_2$ is non-square by hypothesis, the proof of Proposition 2.6 shows that a_{2n} is non-square for all $n \geq 1$. Thus a_k is non-square for all $k \geq 1$ when $c \equiv i \bmod a+bi$.

In the case $c \equiv -i \bmod a+bi$ we have

$$\{a_n(c)\}_{n\geq 1} \equiv \{1, 1-i, -i, -1+i, -i, -1+i, ...\} \bmod a+bi,$$

so the claim is still true following the above proof.

Since $c \equiv \pm i \bmod a+bi$ by hypothesis, c is non-square mod $a+bi$, so c is non-square in $\mathbb{Z}[i]$. Thus f_r is irreducible. Since $\{a_n\}_{n\geq 2}$ contains no squares in all cases, f_r^n is irreducible for all $n \geq 1$ by Lemma 2.2.

□

Remark. *This theorem can be used to show that if $c \in \mathbb{Z}$ with $c \equiv 2 \bmod 4$, then $f_{1/c}^n$ is irreducible for all n. Note $c \equiv -i \bmod c+i$ and $N(c+i) = c^2+1 \equiv 5 \mod 8$. The hypothesis that $c+1$ is non-square is satified since $c+1 \equiv 3 \mod 4$, but squares in $\mathbb{Z}$ are 0 or $1 \bmod 4$.*

3.0.3. *Case (3).*

Lemma 3.5. *If $c \in \mathbb{Z}[i]\backslash\{0, -1, -1\pm 2i\}$ and $v_\pi(c)$ is odd for all primes π dividing c, then f_r^n is irreducible for all $n \geq 1$.*

Proof. By Table 1, we may assume $c \in \{0, 1\} \bmod 2$. Also, note that $c \neq \alpha^2(2i-\alpha^2)$ unless α is a unit in $\mathbb{Z}[i]$ with $c = -1 \pm 2i$, contrary to our hypothesis. We may thus assume that f_r^2 is irreducible. We proceed by induction on n, assuming f_r^{n-1} is irreducible to prove that f_r^n is irreducible.

We now follow this basic outline: Suppose a_n is a square in $\mathbb{Z}[i]$. We first show $b_n^{\pm} \notin \{0, \pm 1\}$ for all c with $|c| > \sqrt{5}$. We also show $b_n^{\pm} \neq \pm 2i$ unless c divides a power of 2 (we handle this case separately); it will follow that both b_n^+, and b_n^- have a prime divisor of odd multiplicity (i.e. are non-square).

Thus, suppose $a_n(c)$ is a square in $\mathbb{Z}[i]$ (if $a_n(c)$ is non-square in $\mathbb{Z}[i]$ then f_r^n is irreducible by the proof of Lemma 2.2). Then $b_n^{\pm}$ are $\mathbb{Z}[i]$-factors of $c^{2^{n-1}-1}$:

$$a_n - a_{n-1}^2 = c^{2^{n-1}-1} = (\sqrt{a_n} - a_{n-1})(\sqrt{a_n} + a_{n-1}) = i(a_{n-1} - \sqrt{a_n})i(a_{n-1} + \sqrt{a_n}) = b_n^- b_n^+.$$

Also, note that

$$\gcd(b_n^-, b_n^+) = \gcd(i(a_{n-1} - \sqrt{a_n}), i(a_{n-1} + \sqrt{a_n})) \mid \gcd(2ia_{n-1}, 2i\sqrt{a_n}) = 2$$

since $\gcd(a_n, a_{n-1}) = \gcd(c^m, a_{n-1}) = 1$. When $c \equiv 1 \bmod 2$, c is coprime to 2, so $\gcd(b_n^-, b_n^+) = 1$.

We first show that we may assume $b_n^\pm(c) \notin \{0, \pm 1\}$. To show $b_n^\pm \neq 0$, suppose $b_n^\pm = 0$. Then $\sqrt{a_n} = \pm a_{n-1}$, so $a_n = a_{n-1}^2 + c^{2^{n-1}-1} = a_{n-1}^2$. Since $c \neq 0$, this is impossible.

If $b_n^\pm = \pm 1$, then $a_{n-1} \pm \sqrt{a_n} = \pm i$. Then $a_n = -1 \mp 2ia_{n-1} + a_{n-1}^2$, so

$$a_{n-1}^2 + c^{2^{n-1}-1} = -1 \mp 2ia_{n-1} + a_{n-1}^2 \implies c^{2^{n-1}-1} = -1 \pm 2ia_{n-1}(c).$$

Since $a_{n-1} \equiv 1 \bmod c$, we must then have $0 \equiv -1 \pm 2i \bmod c$, so $c = \mu \cdot (-1 \pm 2i)$ for some $\mu \in \mathbb{Z}[i]^\times$ since $-1 \pm 2i$ is prime in $\mathbb{Z}[i]$. However, $c = \pm 2 + i$ and $c = 1 + 2i$ are found in Table 1 as pairs $(c \bmod z, z) = (i, 2)$ and $(1 + 2i, 4)$ respectively, so a_n is non-square for all $n \geq 2$ for these c. Thus $f_{1/(\pm 2+i)}$, and $f_{1/(1+2i)}$ are stable by Lemma 2.2. Note f_r for $c = 1 - 2i$ has the same behavior by Proposition 2.1. Thus we may assume $b_n^\pm \notin \{0, \pm 1\}$.

First suppose $c \equiv 1 \bmod 2$. We've shown that we may assume $b_n^\pm \notin \{0, \pm 1\}$. If say $b_n^+ \in \{-i, i\}$, then b_n^+ is non-square in $\mathbb{Z}[i]$, with $b_n^+ \equiv b_n^- \equiv i \bmod 2$, so $b_n^\pm$ is non-square in $\mathbb{Z}[i]$.

Now suppose $|b_n^\pm| > 1$. Then $b_n^\pm$ has a prime divisor π with $\pi \nmid b_n^\mp$, since $\gcd(b_n^-, b_n^+) = 1$. This prime divisor is of odd multiplicity since $v_\pi(b_n^\pm) = v_\pi(c^m) = m \cdot v_\pi(c)$ and both m and $v_\pi(c)$ are odd by hypothesis. Thus $b_n^\pm$ is non-square.

Now suppose $c \equiv 0 \bmod 2$. We now show $b_n^\pm(c) \neq \pm 2i$ unless $c \mid 8$ or $c \mid 2^{n-1}$. The rest of the proof will be analogous to the $c \equiv 1 \bmod 2$ case above.

Suppose $b_n^\pm(c) = \pm 2i$. Then

$$a_{n-1} \pm \sqrt{a_n} = \pm 2 \implies a_n = 4 \mp 4a_{n-1} + a_{n-1}^2.$$

Therefore $c^{2^{n-1}-1} + a_{n-1}^2 = 4(1 \pm a_{n-1}) + a_{n-1}^2$, so

$$c^{2^{n-1}-1} = 4(1 \pm a_{n-1}).$$

Since $c^{2^{n-1}-1} = 4(1 \pm a_{n-1})$, first suppose $c^{2^{n-1}-1} = 4(a_{n-1} + 1)$. Since $a_{n-1} \equiv 1 \bmod c$, then $4(a_{n-1} + 1) \equiv 8 \equiv c^{2^{n-1}-1} \equiv 0 \bmod c$, so $c \mid 8 = i(1 + i)^6$.

Otherwise, we have $c^{2^{n-1}-1} = 4(1 - a_{n-1})$. Note that the constant term of $\frac{a_{n-1}(c)-1}{c}$ is 2^{n-3} for $n \geq 3$ by induction: when $n = 3$, we have $\frac{a_2(c)-1}{c} = \frac{(c+1)-1}{c} = 1$. Now suppose $k \geq 3$, and $\frac{a_{k-1}(c)-1}{c}$ has a constant term of 2^{k-3} to show $\frac{a_k(c)-1}{c}$ has constant term 2^{k-2}. In particular, by the hypothesis we know

$$a_{k-1}(c) = p_{k-1}(c) + 2^{k-3}c + 1$$

for some polynomial $p_{k-1}(c)$ divisible by c^2. Then

$$\begin{aligned} a_k(c) = c^{2^{k-1}-1} + a_{k-1}^2(c) &= c^{2^{k-1}-1} + (p_{k-1}(c) + 2^{k-3}c + 1)^2 \\ &= c^{2^{k-1}-1} + p_{k-1}^2 + 2p_{k-1} \cdot 2^{k-3}c + 2p_{k-1} + (2^{k-3}c)^2 + 2 \cdot 2^{k-3}c + 1 \\ &= p_k(c) + 2^{k-2}c + 1 \end{aligned}$$

where p_k is a polynomial divisible by c^2 which was the claim to be shown. Then

$$0 \equiv c^{2^{n-1}-2} \equiv 4\frac{1 - a_{n-1}(c)}{c} \equiv 4 \cdot -2^{n-3} \equiv -2^{n-1} \bmod c$$

since the constant term of $\frac{a_{n-1}(c)-1}{c}$ is 2^{n-3}. Therefore $c \mid 2^{n-1}$ when $c^{2^{n-1}-1} = 4(1 - a_{n-1})$.

We now show that if c divides a power of 2 (with $v_{1+i}(c)$ odd), then $b_n^\pm(c)$ is not a square in $\mathbb{Z}[i]$.

Lemma 3.6. *If c divides a power of 2 with $v_{1+i}(c)$ odd, then $b_n^{\pm}(c)$ is not a square in $\mathbb{Z}[i]$.*

Proof. If $v_{1+i}(c)$ is odd but $v_\pi(c) = 0$ for all primes $\pi \neq 1+i$, then c or $\bar{c}$ is of the form $\pm 2^k(1+i)$, both non-square so f_r is irreducible. By Proposition 2.1, we assume $c = \pm 2^k(1+i)$.

It is readily shown that $c+1 = \pm 2^k + 1 \pm (2^k)i$ is non-square when $c \neq -4-4i$. Note that $a_n(-4-4i) \equiv -1-i \bmod 3$. If $c \equiv -1-i \bmod 3$, we obtain the list

$$\{a_n\}_{n\geq 1} \equiv \{1, -i, 1+i, 1+i, ...\} \bmod 3$$

and note that $1+i$ is non-square mod 3. Thus a_n is non-square for $n \geq 3$, so $b_n^{\pm}$ is non-square for all $n \geq 3$.

Now we may assume $a_2 = c+1$ is non-square. Note $c \equiv 0 \bmod 2$ implies we have $c+1 \equiv 1 \bmod 2$. Since a_2 is non-square, it is not of the form $i\alpha^2$ since $i\alpha^2 \in \{0, i\} \bmod 2$, hence a_2 has a prime divisor of odd multiplicity.

Also note that $c = \pm 2^k(1+i) \equiv \pm(1+i) \bmod 3$. When $c \equiv -1-i \bmod 3$, we've shown above that $b_n^{\pm}(c)$ is not a square in $\mathbb{Z}[i]$ for all $n \geq 3$.

From here on assume $c \equiv 1+i \bmod 3$. We show that $b_n^{\pm} \neq \pm 2i$; it will follow that $b_n^{\pm}$ is non-square. Suppose for now that $b_n^{\pm} \neq \pm 2i$ to show this implication. Recall that $a_n = a_{n-1}^2 + c^m$ for $m = 2^{n-1} - 1$. Note $\gcd(b_n^-, b_n^+) \mid 2$ and $c^m = \pm i^{km}(1+i)^{(2k+1)m} = b_n^- b_n^+$ with $b_n^{\pm} \notin \{\pm 1, \pm 2i\}$. Suppose without loss of generality that $v_{1+i}(b_n^+) = 2$. Since $b_n^+ \neq \pm 2i$, we then must have $b_n^+ = \pm 2$ and $v_{1+i}(b_n^-)$ odd (since m and $2k+1$ are odd), so both $b_n^{\pm}$ are non-square.

If $v_{1+i}(b_n^+) = 1$, then $b_n^+ \equiv b_n^- \equiv 1+i \bmod 2$, so both $b_n^{\pm}$ are non-square.

Note that $v_{1+i}(b_n^+) = 0$ is a contradiction since $b_n^- \equiv b_n^+ \equiv 1 \bmod 1+i$ while

$$b_n^- \cdot b_n^+ = c^m \equiv 0 \bmod 1+i.$$

In all cases, $b_n^{\pm}$ is non-square.

Recall that if $b_n^{\pm} = \pm 2i$, we have $c^m = 4(1 \pm a_{n-1}) \implies c \mid 2^{n-1}$ or 8. We first show that the $c \mid 8$ cases pose no obstruction, obtained when $c^m = 4(1 + a_{n-1})$.

If $c = \pm 2^k(1+i)$ divides 8, then $k \in \{1, 2\}$ since by assumption $c \equiv 0 \mod 2$. The values of c also satisfying $c \equiv 1+i \bmod 3$ are $c \in \{-2-2i, 4+4i\}$. When $c = -2-2i$, we have that $c+1 = -1-2i$ is non-square in $\mathbb{Z}[i]$. We find from Table 2 that the modulus $5+i$ ensures a_n is non-square for $n \geq 2$. When $c = 4+4i$, we find from Table 1 the modulus $3+i$. Thus $b_n^{\pm}(c)$ is non-square for all non-square $c \mid 8$ and $n \geq 2$.

When $c^m = 4(1 - a_{n-1})$, we obtained $c \mid 2^{n-1}$. We now show that $b_n^{\pm} = \pm 2i$ and $c \equiv 1+i \mod 3$ leads to a contradiction. Now, we have

$$c^m = 4(1 - a_{n-1}) \implies c^{2^{n-1}-1} = 4(1 - a_{n-2}^2 - c^{2^{n-2}-1}) \tag{6}$$

Note we may assume $n > 4$ since $a_3(c)$ is non-square by Theorem 2.7 and since a_4 is non-square when a_2 is non-square, implying that $b_j^{\pm}$ is non-square in $\mathbb{Z}[i]$ for $j \in \{2, 3, 4\} \cup 2\mathbb{Z}^+$.

Note that

$$\{c^k \bmod 3\}_{k\geq 1} \equiv \{(1+i)^k \bmod 3\}_{k\geq 1} \equiv \{1+i, -i, 1-i, 2, -1-i, i, -1+i, 1, ...\}$$

so $1+i$ and thus c has multiplicative order 8 mod 3. Since $8 \mid 2^k$ when $k \geq 3$, we have

$$c^{2^{n-1}-1} \equiv c^{-1} \equiv -1+i \mod 3.$$

Since $n > 4$, we have $2^{n-1} - 1 \equiv 2^{n-2} - 1 \equiv -1 \bmod 8$ and we obtain the following from equation (6):

$$-1+i \equiv 1 - a_{n-2}^2 - (-1+i) \mod 3$$

We obtain $a_{n-2}^2 \equiv -2i \bmod 3$, implying $a_{n-2} \equiv \pm(1-i) \bmod 3$. Since

$$\{a_n(c) \bmod 3\}_{n\geq 1} \equiv \{1, -1+i, 1, i, 1+i, -1, i, 1+i, -1, ...\},$$

we must have $n-2=2$ in this case, so $n=4$. Since $n>4$ by assumption, this is a contradiction.

Thus $b_n^{\pm} \neq 2i$; we've shown that this implies $b_n^{\pm}$ is non-square. Note that since c is non-square, f_r is irreducible. Since $\{b_n^+(c), b_n^-(c)\}$ contains no squares in $\mathbb{Z}[i]$ for each $n \geq 3$, it follows that $f_{1/c}^n(x)$ is irreducible for all $n \geq 1$ by Lemma 3.1. □

□

Remark. *One may apply Theorem 3.2 to $c=-89$, a case with a_3 square in $\mathbb{Z}[i]$ by Theorem 2.7. Since $-89 \equiv 3 \bmod 4$ is a Gaussian prime, item (3) of this proposition applies (item (1) also applies since $2 \nmid c$ and $16 \nmid c+1=-88 \notin \mathbb{Z}[i]^2$), hence we have $f_{-1/89}^n(x)$ is irreducible for all $n \geq 1$ by Lemma 3.1.*

4. A useful factorization of c

We first note the following extension of [6, Lemma 4.1] to the Gaussian integers (where $N : \mathbb{C} \to \mathbb{R}$ is the field norm $N_{\mathbb{C}/\mathbb{R}}(a+ib) = a^2+b^2$):

Lemma 4.1. *Let $c \in \mathbb{Z}[i]$ with $|c|>1$, and $n \geq 2$ such that $a_n = a_n(c)$ is a square. Then we may write $c=uv$ with $\gcd(u,v)=1$, such that*

$$dv^{2^{n-1}-1} - \frac{u^{2^{n-1}-1}}{d} = 2a_{n-1}(uv)$$

where $d \in \{1, 1+i, 2\}$ (if $n>2$, then $d \neq 1+i$). If in addition $\{b_n^+(c), b_n^-(c)\}$ contains a square, then $\{N(u), N(v)\}$ contains a square.

Proof. This is the proof of [6, Lemma 4.1] with minor modifications. For a sketch of the proof: when $a_n(c)$ is a square, say s^2, then $c^m = (s+a_{n-1}(c))(s-a_{n-1}(c))$ for some $s \in \mathbb{Z}[i]$. Define $d := \gcd(s+a_{n-1}(c), s-a_{n-1}(c))$, $t_+ := s+a_{n-1}(c)$, and $t_- := s-a_{n-1}(c)$, so that $t_+t_- = c^m$ implies $c=uv$ with $t_+ = dv^m$ and $t_- = u^m/d$, or $t_+ = u^m/d$ and $t_- = dv^m$.

Note that $d = 1+i$ is impossible when $n>2$: we have

$$s + a_{n-1}(c) \equiv s - a_{n-1}(c) \equiv 1+i \bmod 2.$$

In this case, we have $v_{1+i}(s \pm a_{n-1}(c)) = 1$. However,

$$v_{1+i}(c^m) = m \cdot v_{1+i}(c) = v_{1+i}(s+a_{n-1}(c)) + v_{1+i}(s-a_{n-1}(c)) = 2,$$

so m odd implies $m = 2^{n-1}-1 = 1$, hence $n=2$, and $v_{1+i}(c)=2$.

Note $b_n^+ = i \cdot t_+$ and $b_n^- = -i \cdot t_-$. If b_n^+ is a square in $\mathbb{Z}[i]$, then $|b_n^+|^2$ is a square in $\mathbb{Z}[i]$. If $d \mid 2$, we have $t_+ = dv^m$ and $t_- = u^m/d$ or $t_+ = u^m/d$ and $t_- = dv^m$. Then

$$|b_n^+|^2 = |t_+|^2 \in \left\{ |dv^m|^2, \left|\frac{u^m}{d}\right|^2 \right\} = \left\{ |d|^2(|v|^2)^m, \frac{(|u|^2)^m}{|d|^2} \right\}.$$

When $n>2$ we may assume $d \in \{1,2\}$, so one of $\{|u|^2, |v|^2\}$ is square in $\mathbb{Z}[i]$ since m is odd. The case when b_n^- is square is similar.

□

We extend [6, Definition 4.3] over $\mathbb{Z}[i]$:

Definition 4.2. *Let $c \in \mathbb{Z}[i]$ with $|c|>1$. We set*

$$Q(c) := \min\{N(v/u) : u, v \in \mathbb{Z}[i] \text{ coprime with } |v|>|u| \text{ and } c=uv\}$$

$$\tilde{Q}(c) := \min\{N(v/u) : u, v \in \mathbb{Z}[i] \text{ coprime with } |v|>|u|, c=uv, \text{ and } \{N(v), N(u)\} \text{ contains a square}\}$$

and

$$q(c) := \sqrt{Q(c)}, \quad \tilde{q}(c) := \sqrt{\tilde{Q}(c)}.$$

If $Q(c) = N(v/u)$, then since $N(v) > N(u)$, $N(v) \geq N(u) + 1$, thus $N(v/u) \geq 1 + 1/N(u)$. Note $N(u) < N(\sqrt{c})$, since $N(u^2) < N(u)N(v) = N(c)$, so

$$\tilde{Q}(c) \geq Q(c) = N\left(\frac{v}{u}\right) > 1 + \frac{1}{N(\sqrt{c})} = 1 + \frac{1}{|c|} \quad \text{and} \quad \tilde{q}(c) \geq q(c) > \sqrt{1 + \frac{1}{|c|}}.$$

We also make use of the following $\mathbb{Z}[i]$ extension of [6, Definition 4.4]:

Definition 4.3. *Let $c \in \mathbb{Z}[i]\backslash\{0, -1\}$ and $n \geq 2$. Define $\epsilon(n, c)$ so that*

$$\log \frac{\sqrt{a_n(c)} + a_{n-1}(c)}{\sqrt{a_n(c)} - a_{n-1}(c)} = \frac{\epsilon(n, c)}{\sqrt{c}}.$$

Suppose $|g(z)| < 1$. As in [12], the ML inequality (or using Maclaurin series) yields

$$\left|\log \frac{1 + g(z)}{1 - g(z)}\right| = \left|2 \int_0^{g(z)} \frac{d\gamma}{1 - \gamma^2}\right| \leq 2 \int_0^{|g(z)|} \frac{dt}{1 - t^2} = \log \frac{1 + |g(z)|}{1 - |g(z)|}. \tag{7}$$

This inequality prompts us to consider the ratio $\frac{a_{n-1}(c)}{\sqrt{a_n(c)}}$. We also consider the related ratio $\frac{a_n(c)}{c^{2^{n-1}-1}}$, which converges nicely when $c \in \mathbb{R}$, as shown by the following lemma. Assume $t \geq 4$ in $\mathbb{R}$, and let $F(t) = \frac{1}{2}(1 - \sqrt{1 - \frac{4}{t}}) = \frac{2}{t(1+\sqrt{1-\frac{4}{t}})}$. Note $F(t) \leq 1/2$, and that $F(t)$ decreases monotonically to 0.

Lemma 4.4. *(Demark-Hindes-Jones-Misplon-Stoll-Stoneman,* [6, Lemma 4.2]*)*
Let $t \geq 4$ in $\mathbb{R}$. Then the sequence

$$\bar{a}_n(t) = \frac{a_n(t)}{t^{2^{n-1}-1}} \quad \text{for } n \geq 1$$

satisfies

$$1 = \bar{a}_1(t) < \bar{a}_2(t) < \ldots \quad \text{and} \quad \lim_{n\to\infty} \bar{a}_n(t) = tF(t).$$

We now seek some condition on $g(z) := \frac{a_{n-1}}{\sqrt{a_n}}(z)$ to guarantee $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| < 1$:

Theorem 4.5. *Let $n \geq 2$. If $|c| \geq 5$, then $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| < 1$.*

Proof. Let $m = 2^{n-1} - 1$. We consider the contrapositive. When $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| \geq 1$, we have $|a_n(c)/a_{n-1}^2(c)| \leq 1$, thus $|a_n(c)| \leq |a_{n-1}^2(c)|$. By the reverse triangle inequality and the relationship $a_n(c) = c^m + a_{n-1}^2(c)$,

$$|c|^m - |a_{n-1}^2(c)| \leq |a_{n-1}^2(c) + c^m| = |a_n(c)| \leq |a_{n-1}^2(c)|$$

implies $|c|^m \leq 2|a_{n-1}^2(c)|$. Therefore $|c|^m \leq 2a_{n-1}^2(|c|)$ since $a_{n-1}(c)$ has non-negative coefficients, so

$$1/2 \leq \frac{a_{n-1}^2(|c|)}{|c|^m} = \frac{a_n(|c|)}{|c|^m} - 1$$

since $a_n(|c|) = a_{n-1}^2(|c|) + |c|^m$. Thus

$$3/2 \leq \frac{a_n(|c|)}{|c|^m} = \bar{a}_n(|c|).$$

Since $5 \cdot F(5) \approx 1.38 < 3/2$, we have that all $|c| = t \in [5, \infty)$ have $\bar{a}_n(t) < tF(t) < 3/2$ since $tF(t)$ decreases on $[4, \infty)$:

$$\frac{\mathrm{d}}{\mathrm{d}t}(tF(t)) = F(t) - t \cdot \left(\frac{1}{t^2\sqrt{1 - 4/t}}\right) = F(t) - \frac{1}{t\sqrt{1 - 4/t}}.$$

Since $F(t) = \frac{2}{t(1+\sqrt{1-\frac{4}{t}})} = \frac{2}{t+t\sqrt{1-4/t}} < \frac{2}{2t\sqrt{1-4/t}} = \frac{1}{t\sqrt{1-4/t}}$ for $t \in (4, \infty)$, we have $\frac{\mathrm{d}}{\mathrm{d}t}(tF(t)) < 0$ on $(4, \infty)$. Thus $tF(t)$ decreases to 1 on this interval.

Our hypothesis $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| \geq 1$ is thus false on $[5, \infty)$. Thus if $|c| \geq 5$, then $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| < 1$. □

Remark. *Note that $c \in \mathbb{Z}[i]$ with $|c| < 5$ implies $|c| \leq 2\sqrt{5}$, while*

$$2\sqrt{5} \cdot F(2\sqrt{5}) \approx 1.51 > 3/2,$$

so 5 is the minimum modulus (for Gaussian integers) for which we can guarantee $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| < 1$ with this method.

When $|c| \geq 5$, then $|\frac{a_{n-1}}{\sqrt{a_n}}(c)| < 1$, hence

$$|\epsilon(n,c)| = \left|\sqrt{c}\log\frac{\sqrt{a_n(c)}+a_{n-1}(c)}{\sqrt{a_n(c)}-a_{n-1}(c)}\right| = \left|\sqrt{c}\log\frac{1+\frac{a_{n-1}}{\sqrt{a_n}}(c)}{1-\frac{a_{n-1}}{\sqrt{a_n}}(c)}\right| \leq \sqrt{|c|}\log\frac{1+|\frac{a_{n-1}}{\sqrt{a_n}}(c)|}{1-|\frac{a_{n-1}}{\sqrt{a_n}}(c)|}$$

Let $m = 2^{n-1}-1$. Using the relation $a_n(c) - c^m = a_{n-1}^2(c)$, it follows that $\frac{a_{n-1}}{\sqrt{a_n}}(c) = \sqrt{a_{n-1}^2/a_n} = \sqrt{1-c^m/a_n}$. Since $\bar{a}_n(c) = \frac{a_n(c)}{c^m}$ and $\bar{a}_{n-1}^2/c + 1 = \bar{a}_n$ (as in the proof of Lemma 4.4), we have that

$$\left|\frac{a_{n-1}}{\sqrt{a_n}}(c)\right| = \left|\sqrt{1-\frac{1}{\bar{a}_n(c)}}\right| = \left|\sqrt{\frac{\bar{a}_n(c)-1}{\bar{a}_n(c)}}\right| = \left|\sqrt{\frac{\bar{a}_{n-1}^2}{c\bar{a}_n}}\right| = \left|\frac{\bar{a}_{n-1}}{\sqrt{c\bar{a}_n}}\right|.$$

By the triangle inequality and Lemma 4.4,

$$|\bar{a}_n(c)| = \left|\frac{a_n(c)}{c^m}\right| \leq \frac{a_n(|c|)}{|c|^m} = \bar{a}_n(|c|) \leq |c|F(|c|).$$

Also, $|c| \geq 5$ implies the inequality $|a_{n-1}(c)/\sqrt{a_n(c)}| < 1 \implies |a_{n-1}^2(c)| < |a_n(c)|$ by Theorem 4.5. By the reverse triangle inequality, $|c^m| - |a_n(c)| \leq |a_n(c) - c^m| = |a_{n-1}^2(c)| < |a_n(c)|$ implies $|c^m| < 2|a_n(c)|$, so

$$\left|\frac{c^m}{a_n(c)}\right| = \left|\frac{1}{\bar{a}_n(c)}\right| < 2 \implies \left|\frac{1}{\sqrt{\bar{a}_n}}\right| < \sqrt{2}.$$

By the previous inequalities,

$$\left|\frac{\bar{a}_{n-1}}{\sqrt{c\bar{a}_n}}\right| < \frac{|c|F(|c|)}{\sqrt{|c|}}\sqrt{2} = F(|c|)\sqrt{2|c|}.$$

Note $F(|c|)\sqrt{2|c|} = \frac{2\sqrt{2}}{\sqrt{|c|}(1+\sqrt{1-4/|c|})}$ decreases as a function of $|c|$ when $|c| \geq 4$. Since $|c| \geq 5$, we have

$$F(|c|)\sqrt{2|c|} \leq F(5)\sqrt{10} \approx 0.87.$$

Since $\log\frac{1+x}{1-x}$ is an increasing function of x on the interval $(-1,1)$,

$$\sqrt{|c|}\log\frac{1+|a_{n-1}(c)/\sqrt{a_n(c)}|}{1-|a_{n-1}(c)/\sqrt{a_n(c)}|} = \sqrt{|c|}\log\frac{1+|\bar{a}_{n-1}|/|\sqrt{c\bar{a}_n}|}{1-|\bar{a}_{n-1}|/|\sqrt{c\bar{a}_n}|} < \sqrt{|c|}\log\frac{1+F(|c|)\sqrt{2|c|}}{1-F(|c|)\sqrt{2|c|}}.$$

This bound leads to the following definition.

Definition 4.6. *Let $|c| \geq 5$, with $c \in \mathbb{Z}[i]$ and $n \geq 2$. Define $\Xi(|c|)$ as*

$$\Xi(|c|) = \sqrt{|c|}\log\frac{1+F(|c|)\sqrt{2|c|}}{1-F(|c|)\sqrt{2|c|}}.$$

We have shown that $|\epsilon(n,c)| < \Xi(|c|)$. It follows from the definition of $F(c)$ (and L'Hopital's rule) that $\Xi(|c|)$ decreases to $\sqrt{2}$ as $|c| \to \infty$. We also have that $\Xi(|c|) \leq \Xi(5) \approx 6.05$.

5. The proof of Theorem 1.9

We now have the tools in place to prove the $\mathbb{Z}[i]$ analogue of [6, Proposition 4.5]:

Proposition 5.1. *Suppose $|c| \geq 5$ and $n \geq 5$. If $c \in \mathbb{Z}[i]\backslash\mathbb{Z}^-$ and*

$$n \geq 1 + \log_2\left(1 + \frac{|\epsilon(n,c)/\sqrt{c}|}{\log q(c)} + \frac{\log 4}{\log q(c)}\right)$$

then $a_n(c)$ is not a square.

If $c \in \mathbb{Z}[i]\backslash\mathbb{Z}^-$, and the weaker condition

$$n \geq 1 + \log_2\left(1 + \frac{|\epsilon(n,c)/\sqrt{c}|}{\log \tilde{q}(c)} + \frac{\log 4}{\log \tilde{q}(c)}\right)$$

holds, then $b_n^{\pm}(c)$ is non-square.

Proof. In the following, we write a_n for $a_n(c)$ since c is fixed, and assume that a_n is a square. Let $m = 2^{n-1}-1$. Since $n \geq 2$, by Lemma 4.1 we may write $c = uv$ with coprime u, v such that

$$(v^m, u^m) = (\frac{1}{d}(\pm\sqrt{a_n} + a_{n-1}), d(\pm\sqrt{a_n} - a_{n-1})) \quad \text{for } d \in \{1, 2\}.$$

We first show $|u| \neq |v|$ in all cases. Note $|u| \neq |v| \iff |u^m| \neq |v^m|$. Let $\sqrt{a_n} = A + Bi$, $a_{n-1} = C + Di$.

When $d = 1$, $N(u^m) = N(v^m)$ implies

$$\begin{aligned} N(\pm\sqrt{a_n} - a_{n-1}) = N(\pm\sqrt{a_n} + a_{n+1}) &\iff N(A + Bi - (C + Di)) = N(A + Bi + (C + Di)) \\ &\iff (A - C)^2 + (B - D)^2 = (A + C)^2 + (B + D)^2 \\ &\iff -2AC - 2BD = 2AC + 2BD \\ &\iff 0 = 4(AC + BD), \end{aligned}$$

so the vectors $\overrightarrow{\sqrt{a_n}} = \begin{bmatrix} A \\ B \end{bmatrix}$ and $\vec{a}_{n-1} = \begin{bmatrix} C \\ D \end{bmatrix}$ are perpendicular.

Then, we must have $\overrightarrow{\sqrt{a_n}} = \lambda \begin{bmatrix} -D \\ C \end{bmatrix}$ for some real λ, i.e.

$$\sqrt{a_n} = \lambda(-D + iC) = \lambda i(C + Di) = \lambda i a_{n-1}.$$

Then $A = -\lambda D$, $B = \lambda C$. Since a_n is a square, $A, B \in \mathbb{Z}$, so λ is a rational number $s/t \in \mathbb{Q}$. To have $\sqrt{a_n} \in \mathbb{Z}[i]$ we must have $t \mid a_{n-1}$. Let $\beta = a_{n-1}/t \in \mathbb{Z}[i]$, so $\beta \mid a_{n-1}$. Then

$$\sqrt{a_n} = (s/t)i a_{n-1} = si\beta.$$

However, a_n is relatively prime to a_{n-1} as shown in the proof of Lemma 4.1. Since $\beta \mid \gcd(\sqrt{a_n}, a_{n-1}) = 1$, we have $\beta \in \{\pm 1, \pm i\}$. Thus $\sqrt{a_n}$ is either real or pure imaginary. If $\beta = \pm 1 = a_{n-1}/t$, then $a_{n-1} \in \mathbb{Z}$, and thus $\sqrt{a_n} = \lambda i a_{n-1} \in i\mathbb{Z}$. The reverse is true for $\beta = \pm i$. In either case, $a_n \in \mathbb{Z}$.

Since $a_n = a_{n-1}^2 + c^m$ and $a_n, a_{n-1}^2 \in \mathbb{Z}$, then $c^m \in \mathbb{Z}$. Since $m = 2^{n-1} - 1$ is odd, we must have $c \in \mathbb{Z}$ (as in [9]). This contradicts a_{n-1} being pure imaginary, since $a_k(c) = a_{k-1}^2 + c^{2^{k-1}-1}, a_1(c) = 1$, and thus $a_k \in \mathbb{Z}$ whenever $c \in \mathbb{Z}$. Thus $\sqrt{a_n}$ is pure imaginary and $c \in \mathbb{Z}$ with $\beta = \pm 1$. The only way $\sqrt{a_n}$ is pure imaginary and $c \in \mathbb{Z}$ is when $a_n(c)$ is a negative integer. Then c is a negative integer, since $a_n(c) > 0$ when $c \in \mathbb{Z}^+$. In this case $A = D = 0$, with

$$N(\sqrt{a_n} - a_{n-1}) = N(\sqrt{a_n} + a_{n-1}) \iff N(iB - C) = N(iB + C).$$

Since $c \in \mathbb{Z}[i]$ is not a negative integer by hypothesis, we must have $N(u) \neq N(v)$.

When $d = 2$, $N(u^m) = N(v^m)$ implies $|\frac{\sqrt{a_n}+a_{n-1}}{\sqrt{a_n}-a_{n-1}}| = 4$. Then $N(\frac{\sqrt{a_n}+a_{n-1}}{\sqrt{a_n}-a_{n-1}}) = 16$:

$$N(\sqrt{a_n} + a_{n-1}) = 16 \cdot N(\sqrt{a_n} - a_{n-1}). \tag{8}$$

Since $d = \gcd(\sqrt{a_n} + a_{n-1}, \sqrt{a_n} - a_{n-1}) = 2$, we must have $v_{1+i}(\sqrt{a_n} - a_{n-1}) = 2$, hence in $\mathbb{Z}$,

$$v_2(N(\sqrt{a_n} - a_{n-1})) = 2.$$

Now $v_2(N(\sqrt{a_n}+a_{n-1}))=6$ in $\mathbb{Z}$ by equation (8). Since $(\sqrt{a_n}+a_{n-1})(\sqrt{a_n}-a_{n-1})=c^m$,

$$8=v_2(N(\sqrt{a_n}+a_{n-1})N(\sqrt{a_n}-a_{n-1}))=v_2(N(c^m))=m\cdot v_{1+i}(c),$$

so $m\mid 8$. This is a contradiction; since $n>2$, $m=2^{n-1}-1$ is an odd number greater than 1. Thus $|v|\neq|u|$ in all cases.

Now, since $|v|\neq|u|$,

$$\begin{aligned} m\log q(c)\leq m|\log(|v/u|)|=|\log(|(v/u)^m|)| &= \left|\log\left|d^{\pm 2}\frac{\sqrt{a_n}+a_{n-1}}{\sqrt{a_n}-a_{n-1}}\right|\right| \\ &= \left|\log|d^{\pm 2}|+\log\left|\frac{\sqrt{a_n}+a_{n-1}}{\sqrt{a_n}-a_{n-1}}\right|\right| \\ &\leq \log 4+\left|\log\frac{\sqrt{a_n}+a_{n-1}}{\sqrt{a_n}-a_{n-1}}\right| \\ &= \log 4+|\epsilon(n,c)/\sqrt{c}|. \end{aligned}$$

This is equivalent to the inequalities shown, since $m=2^{n-1}-1$.

Assuming that $b_n^{\pm}(c)$ is square in $\mathbb{Z}[i]$ yields $N(u)$ or $N(v)$ is square by Lemma 4.1, hence the bound is valid for $\tilde{q}(c)$ in place of $q(c)$.

□

Remark. *Note that* [6, Proposition 4.5] *assumed c is a non-square positive integer $c=uv$, so $u\neq\pm v$ and thus $|u|\neq|v|$ is automatic.*

We now handle the negative integer case. We'll need [6, Proposition 3.1] as a lemma:

Lemma 5.2. *(Demark-Hindes-Jones-Misplon-Stoll-Stoneman,*[6, Proposition 3.1]*) If $c\in\mathbb{Z}^-$, then $a_n\in\mathbb{Z}^-$ for all $n\geq 2$.*

We can now prove that if $c\in\mathbb{Z}^-$ is non-square in $\mathbb{Z}[i]$, then $b_n^{\pm}$ is not a square in $\mathbb{Z}[i]$:

Theorem 5.3. *Suppose $c\in\mathbb{Z}^-\backslash\mathbb{Z}[i]^2$ is a negative integer. Then $f_r^n(x)$ is irreducible for all $n\geq 1$.*

Proof. Since c is a negative integer, $a_n(c)\in\mathbb{Z}^-$ by Lemma 5.2. Thus, if $a_n(c)$ is a square in $\mathbb{Z}[i]$, $\sqrt{a_n(c)}\in i\mathbb{Z}$. If $b_n^{\pm}(c)$ is a square in $\mathbb{Z}[i]$ for some $n>2$, then by the proof of Proposition 5.1 we may write $c=uv$ with $\gcd(u,v)=1$ such that $\{N(u),N(v)\}$ contains a square, and with

$$(v^m,u^m)=\left(\frac{1}{d}(\pm\sqrt{a_n}-a_{n-1}),d(\pm\sqrt{a_n}+a_{n-1})\right)=\left(\frac{iB-C}{d},d(iB+C)\right)\qquad\text{for } B,C\in\mathbb{Z},\text{ with } d\in\{1,2\}.$$

By multiplicativity of the norm $N(\cdot)=N_{\mathbb{C}/\mathbb{R}}(\cdot)$, we have $N(v^m)=d^4\cdot N(u^m)$. Thus one of $\{N(u),N(v)\}$ being a square forces the other to be a square. Note $|v^m|=d^2|u^m|$. Since $c^m=u^mv^m$, we have $|c^m|=|c|^m=|u^m||v^m|=d^2|u^m|^2=d^2N(u)^m$ is a square in $\mathbb{Z}$. Since m is odd, we must have that $|c|=-c$ is a square in $\mathbb{Z}$ (so in particular, c is a square in $\mathbb{Z}[i]$), contrary to our hypothesis. Thus $b_n^{\pm}(c)$ is not a square in $\mathbb{Z}[i]$ for all $n>2$.

We now show $f_r^2(x)$ is irreducible to apply Lemma 3.1, it will follow that we have f_r^n irreducible over $\mathbb{Z}[i]$ for all $n\geq 1$. If $c\neq\alpha^2(2i-\alpha^2)\in\mathbb{Z}^-\backslash\mathbb{Z}[i]^2$, then f_r^2 is irreducible by Proposition 2.3: suppose the opposite for a contradiction.

If $c=\alpha^2(2i-\alpha^2)$, then $c\in\{-1\pm 2i,0\}$ mod 4. We may assume $c\equiv 0 \bmod 4$ since $c\in\mathbb{Z}^-$. Then $(1+i)\mid\alpha$, so $c=4\eta^2(\eta^2\pm 1)$ for some $\eta\in\mathbb{Z}[i]$. Since $c\in\mathbb{Z}$, we may assume $\eta\in\mathbb{Z}\cup i\mathbb{Z}$: if $\eta=A+Bi$, then

$$0=\mathrm{Im}(c)=\mathrm{Im}(4\eta^2(\eta^2\pm 1))=16A^3B-16AB^3\pm 8AB.$$

If both A,B are non-zero, then $16A^2-16B^2\pm 8=0$, so $\pm 8=16(B^2-A^2)$. This is a contradiction, so we may assume A or $B=0$.

Thus $\eta\in\mathbb{Z}\cup i\mathbb{Z}$. If $|\eta|>1$ in $\mathbb{Z}$ or $i\mathbb{Z}$ we have $c=4\eta^2(\eta^2\pm 1)\in\mathbb{Z}^+$. If $|\eta|=1$, we have $c\in\{0,8\}$. Thus $c=\alpha^2(2i-\alpha^2)\in\mathbb{Z}^-$ is a contradiction, so $c\neq\alpha^2(2i-\alpha^2)$.

□

The lower bound on n from Proposition 5.1 leads to the following $\mathbb{Z}[i]$ analogue of [6, Corollary 4.6]:

Corollary 5.4. *Let $f(x) = x^2 + 1/c$ for $c \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$ with $|c| \geq 5$.*

(1) *If f^k is irreducible for*

$$k = 1 + \left\lfloor \log_2 \left(1 + \frac{\log 4 + \Xi(|c|)/\sqrt{|c|}}{\log \sqrt{1 + 1/|c|}} \right) \right\rfloor,$$

then all f^n are irreducible.

(2) *If $a_2(c) = c + 1$ and $a_p(c)$ are non-square for all prime numbers p with*

$$5 \leq p \leq 1 + \left\lfloor \log_2 \left(1 + \frac{\log 4 + \Xi(|c|)/\sqrt{|c|}}{\log \sqrt{1 + 1/|c|}} \right) \right\rfloor,$$

then all f^n are irreducible.

Proof. This is the proof [6, Corollary 4.6] using $|\epsilon(n, c)| \leq \Xi(|c|)$ instead of $\epsilon(n, c) \leq \epsilon(c)$. □

The $\mathbb{Z}[i]$ analogue of [6, Proposition 5.4] is again proved with minor modifications:

Proposition 5.5. *Let $c \in \mathbb{Z}[i]$ and set $f(x) = x^2 + 1/c$. If $N(c) \leq 10^9$ and the second iterate f^2 is irreducible, then all iterates of f are irreducible.*

Proof. We consider the set of non-squares not of the form $\alpha^2(2i - \alpha^2)$: this is a necessary and sufficient condition for f_r^2 to be irreducible by Proposition 2.3. When $|c| < 5$, the only cases not handled by Theorem 3.2 are $c \in \pm\{4i, 2 + 4i, 4 + 2i\}$, but $c = -4i$ has the same behavior as $4i$ by Proposition 2.1.

Note that when $c \in \{4i, \pm(4 + 2i), -2 - 4i\}$ we have that $c + 1$ is prime in $\mathbb{Z}[i]$. Hence Table 2 applies to these c with the following congruence pairs (c, z): $(-i, 2 + i)$ for $c = 4i \equiv -i \bmod 2 + i$, $(\pm(1 + i), 3 + i)$ for $c = \pm(4 + 2i) \equiv \pm(1 + i) \bmod 3 + i$, and $(-1 + i, 3 + 2i)$ for $c = -2 - 4i \equiv -1 + i \bmod 3 + 2i$. Thus $a_n(c)$ is non-square in $\mathbb{Z}[i]$ for all $n \geq 2$ for these c.

This leaves $c = 2 + 4i$, with $c + 1 = (2 + i)^2$ a square. However $\{a_n\}_{n\geq 3} = \{1 - i, 1 - i, ...\} \bmod 3$. Since $1 - i$ is non-square mod 3, a_n is non-square for all $n \geq 3$. Since f^2 is irreducible by hypothesis, f^n is irreducible for all $n \geq 1$ by Lemma 3.1.

We now assume $|c| \geq 5$. By Theorem 3.2 it suffices to consider $2 \mid c$ or $16 \mid c + 1$, with $\pm c$ in the first quadrant by Corollary 2.1. In either case, when $a_2(c)$, $a_p(c)$ are non-square for integer primes

$$5 \leq p \leq 1 + \left\lfloor \log_2 \left(1 + \frac{\log 4 + \Xi(10^{4.5})/\sqrt{10^{4.5}}}{\log \sqrt{1 + 1/10^{4.5}}} \right) \right\rfloor = 17 \tag{9}$$

then all f^n are irreducible by Corollary 5.4. This is a finite computation. □

6. Applications to the density of primes dividing orbits

Let K be a number field with ring of integers $\mathcal{O}_K$. Let $S \subset \mathcal{O}_K$ be a set of prime ideals, $\mathfrak{p} \in S$, and denote by $N(\mathfrak{p}) = \#\{\mathcal{O}_K/\mathfrak{p}\mathcal{O}_K\}$ the absolute norm of the ideal $\mathfrak{p}$. We denote by $D(S)$ the Dirichlet density of the set S:

$$D(S) = \lim_{s\to 1^+} \frac{\sum_{\mathfrak{p}\in S} N(\mathfrak{p})^{-s}}{\sum_{\mathfrak{p}} N(\mathfrak{p})^{-s}}$$

Since K is a number field, this density can be shown to be equivalent to the the natural density of S, denoted by

$$\delta(S) = \lim_{B\to\infty} \frac{\#\{\mathfrak{p} \in S : N(\mathfrak{p}) \leq B\}}{\#\{\mathfrak{p} : N(\mathfrak{p}) \leq B\}}.$$

The following is an extension to $\mathbb{Z}[i]$ of [6, Proposition 6.1], requiring minor modifications:

Theorem 6.1. *Let $c, c+1 \in \mathbb{Z}[i]\backslash\mathbb{Z}[i]^2$, $r = 1/c$, and assume that $\{|b_n^+|^2, |b_n^-|^2\}$ contains no squares in $\mathbb{Z}$ for all $n \geq 3$. Then for any $t \in \mathbb{Q}(i)$, we have*

$$D(\{\pi \text{ prime in } \mathbb{Z}[i] : \pi \text{ divides } O_{f_r}(t)\}) = 0.$$

Proof. Let $H := \mathbb{Q}(i)(\sqrt{r}, \sqrt{\bar{r}})$, and note $H_1 := \mathbb{Q}(i)(\sqrt{r})$ (from the proof of Lemma 3.1) is a subfield. Now $f_r = (x + i\sqrt{r})(x - i\sqrt{r})$ over H or H_1. Let $g_1 = (x + i\sqrt{r})$, $g_2 = (x - i\sqrt{r})$. We'll show $g_i(f^{n-1}(0))$ is non-square in H for all $n \geq 2$. Note $g_i(f_r^{n-1}(0)) = f_r^{n-1}(0) \pm i\sqrt{r} \in K_1$, and that following the proof of Lemma 3.1, $f_r^{n-1}(0) \pm i\sqrt{r}$ is a square in H_1 if and only if $i(f_r^{n-1}(0) \pm \sqrt{f_r^n(0)})$ is a square in $\mathbb{Q}(i)$, which is equivalent to $b_n^{\pm} = i(a_{n-1} \pm \sqrt{a_n})$ being a square in $\mathbb{Z}[i]$. By assumption, $b_n^{\pm}$ is non-square in $\mathbb{Z}[i]$ for any $n \geq 3$, so $g_i(f_r^{n-1}(0))$ is not a square in H_1. If it were square in H, then there exist $a, b \in H_1$ such that $g_i(f_r^{n-1}(0)) = (a + b\sqrt{\bar{r}})^2$ with $b \neq 0$. Then $g_i(f_r^{n-1}(0)) = f_r^{n-1}(0) \pm i\sqrt{r}$ can be written as

$$a^2 + \bar{r}b^2 + 2ab\sqrt{\bar{r}} = f_r^{n-1}(0) \pm i\sqrt{r}.$$

In particular, $2ab = 0$ otherwise $2ab\sqrt{\bar{r}} \notin H_1$ can be written as $f_r^{n-1}(0) \pm i\sqrt{r} - (a^2 + \bar{r}b^2) \in H_1$, so $a = 0$ since $b \neq 0$ by assumption. Thus

$$f_r^{n-1}(0) \pm i\sqrt{r} = \bar{r} \cdot b^2 \in \mathbb{Q}(i)$$

is a contradiction since $f_r^{n-1}(0) \pm i\sqrt{r} \in H_1\backslash\mathbb{Q}(i)$. Thus $f_r^{n-1}(0) \pm i\sqrt{r}$ is not a square in H. Since $b \in H_1$, $b = x + yi\sqrt{r}$ for some $x, y \in \mathbb{Q}(i)$, so $f_r^{n-1}(0) \pm i\sqrt{r} = \bar{r} \cdot (x^2 - ry^2 + 2xyi\sqrt{r}) = \bar{r}(x^2 - ry^2) + 2xyi\bar{r}\sqrt{r}$, so $2xy\bar{r} = \pm 1$ implies $y = \frac{\pm 1}{2x\bar{r}}$. Therefore

$$f_r^{n-1}(0) = \bar{r}(x^2 - r(\frac{\pm 1}{2x\bar{r}})^2) \implies \bar{r}x^4 - f_r^{n-1}(0)x^2 - \frac{r}{4\bar{r}} = 0.$$

Thus $x^2 = \frac{f_r^{n-1}(0) \pm \sqrt{f_r^n(0)}}{2\bar{r}} = \frac{a_{n-1}(c) \pm \sqrt{a_n(c)}}{2c^{2^{n-1}}\bar{r}}$, so $f_r^{n-1}(0) \pm i\sqrt{r}$ is a square in H if and only if $\frac{f_r^{n-1}(0) \pm \sqrt{f_r^n(0)}}{2\bar{r}}$ is a square in $\mathbb{Q}(i)$.

Observe that

$$\begin{aligned}
N_{\mathbb{Q}(i)/\mathbb{Q}}(x^2) = \overline{x^2} \cdot x^2 &= \overline{(\frac{a_{n-1}(c) \pm \sqrt{a_n(c)}}{2c^{2^{n-1}}\bar{r}})} \cdot \frac{(a_{n-1}(c) \pm \sqrt{a_n(c)})}{2c^{2^{n-1}}\bar{r}} \\
&= \frac{N_{\mathbb{Q}(i)/\mathbb{Q}}(a_{n-1}(c) \pm \sqrt{a_n(c)})}{4N_{\mathbb{Q}(i)/\mathbb{Q}}(c^{2^{n-1}})N_{\mathbb{Q}(i)/\mathbb{Q}}(r)} \\
&= \frac{N_{\mathbb{Q}(i)/\mathbb{Q}}(a_{n-1}(c) \pm \sqrt{a_n(c)})}{4N_{\mathbb{Q}(i)/\mathbb{Q}}(c^{2^{n-1}})N_{\mathbb{Q}(i)/\mathbb{Q}}(\frac{1}{c})} \\
&= \frac{N_{\mathbb{Q}(i)/\mathbb{Q}}(a_{n-1}(c) \pm \sqrt{a_n(c)})}{4N_{\mathbb{Q}(i)/\mathbb{Q}}(c^{2^{n-1}-1})} \\
&= \frac{1}{4N_{\mathbb{Q}(i)/\mathbb{Q}}(a_{n-1}(c) \mp \sqrt{a_n(c)})}.
\end{aligned}$$

By hypothesis, neither of $N_{\mathbb{Q}(i)/\mathbb{Q}}(b_n^{\pm}(c)) = |b_n^{\pm}|^2 = N_{\mathbb{Q}(i)/\mathbb{Q}}(a_{n-1}(c) \pm \sqrt{a_n(c)}))$ are a square in $\mathbb{Z}$. This is a contradiction since $N_{\mathbb{Q}(i)/\mathbb{Q}}(x^2)$ is a square in $\mathbb{Q}$. Thus $g_i(f_r^{n-1}(0))$ is not a square in H. Note also that $-g_i(f(0)) = -r \mp \sqrt{-r}$ is a square in $H \iff \frac{-r \mp \sqrt{r^2+r}}{2\bar{r}} = \frac{-1 \mp \sqrt{c+1}}{2c\bar{r}}$ is a square in $\mathbb{Q}(i)$, but $c+1$ is non-square in $\mathbb{Q}(i)$ by hypothesis.

Thus we may apply part (2) of [7, Theorem 1.1] twice to show

$$0 = \lim_{B \to \infty} \frac{\#\{\mathfrak{p} \in S : N(\mathfrak{p}) \leq B\}}{\#\{\mathfrak{p} : N(\mathfrak{p}) \leq B\}} \tag{10}$$

where $N(\mathfrak{p}) = \#(\mathcal{O}_H/\mathfrak{p}\mathcal{O}_H)$ is the absolute norm of the ideal $\mathfrak{p}$ and S is the set of primes $\mathfrak{p}$ in the ring of integers $\mathcal{O}_H$ of K that divide $g_i(f_r^{n-1}(t))$ for at least one $i \in \{1, 2\}$ and at least one $n \geq 2$.

Note $H = \mathbb{Q}(i)(\sqrt{r}, \sqrt{\bar{r}})$ is a Galois extension of $\mathbb{Q}$ of degree 8. Excluding ramified primes $\mathfrak{p} \in \mathcal{O}_H$ (this is a finite set of primes), $\mathfrak{p} \in \mathcal{O}_H$ either has absolute norm $p \in \mathbb{Z}$ (when p splits in $\mathcal{O}_H$), or absolute norm p^k

for $2 \leq k \leq 8$ otherwise. We may thus assume that $N(\mathfrak{p}) = p^k$ for some $1 \leq k \leq 8$.

Note that for $k \geq 2$, $\#\{n \leq B : n = p^k \text{ for some prime } p\}$ has asymptotic density zero relative to $\#\{n \leq B : n = p \text{ for some prime } p\}$ and so equation (10) is equivalent to

$$0 = \lim_{B \to \infty} \frac{\#\{\mathfrak{p} \in S : N(\mathfrak{p}) = p \leq B\}}{\#\{\mathfrak{p} : N(\mathfrak{p}) = p \leq B\}} \tag{11}$$

Suppose $\mathfrak{p} \in S$, and say $\mathfrak{p} \mid g_i(f_r^{n-1}(t))$ for some $n \geq 2$. Then

$$N_{H/\mathbb{Q}(i)}(\mathfrak{p}) \mid N_{H/\mathbb{Q}(i)}(g_i(f_r^{n-1}(t))) = \prod_{\sigma \in \mathrm{Gal}(K/\mathbb{Q}(i))} \sigma(g_i(f_r^{n-1}(t))) = (f_r^n(t))^2.$$

Let $(p) = \mathbb{Z} \cap \mathfrak{p}\mathcal{O}_H$, $(\pi) = \mathbb{Z}[i] \cap \mathfrak{p}\mathcal{O}_H$ be the primes lying below $\mathfrak{p}$, and note $\pi \mid N_{H/\mathbb{Q}(i)}(\mathfrak{p})$. Note that $N(\mathfrak{p}) = p$ if and only if p splits in $\mathcal{O}_H$. This is true if and only if π splits in $\mathcal{O}_H$, i.e. $N_{\mathbb{Q}(i)/\mathbb{Q}}(\pi) = p$ and r is a quadratic residue mod π and $\overline{\pi}$. Otherwise, $N(\mathfrak{p}) = p^k$ for some $2 \leq k \leq 8$. But $\pi \mid (f_r^n(t))^2$, so $\pi \mid f_r^n(t)$ and thus $\overline{\pi} \mid \overline{f_r^n(t)} = f_{\overline{r}}^n(\overline{t})$ with

$$0 \equiv f_r(f_r^{n-1}(t)) \equiv (f_r^{n-1}(t))^2 + r \bmod \pi$$

and

$$0 \equiv f_{\overline{r}}(f_{\overline{r}}^{n-1}(\overline{t})) \equiv (f_{\overline{r}}^{n-1}(\overline{t}))^2 + \overline{r} \bmod \overline{\pi}.$$

Thus r must be a quadratic residue mod π, and $\overline{r}$ is a quadratic residue mod $\overline{\pi}$ implies r is one as well. It follows that the numerator of equation (11) is

$$4\#\{\pi \in \mathbb{Z}[i] : N(\pi) = p \leq B \text{ and } \pi \text{ divides } O_f(t)\}.$$

The denominator is

$$4\#\{\pi \in \mathbb{Z}[i] : N(\pi) = p \leq B \text{ and } r \text{ is a quadratic residue mod } \pi \text{ and } \overline{\pi}\}.$$

By Chebotarev's density theorem, the proportion of integer primes that split completely in a Galois extension of degree n is $1/n$, thus the latter is asymptotic to $C \cdot \#\{\pi \in \mathbb{Z}[i] : N(\pi) = p \leq B\}$ for some constant C. It follows that

$$D(\{\pi \text{ prime in } \mathbb{Z}[i] : \pi \text{ divides } O_f(t)\}) = 0$$

as desired. □

Remark. *The hypothesis that $|b_n^{\pm}(c)|^2$ are non-square is strictly weaker than $a_n(c)$ being non-square for $n \geq 2$. When $a_n(c)$ is non-square for all $n \geq 2$, the conclusion follows by (2) of* [7, Theorem 1.1].

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